Light

Data Representation — Media (Text, Images, Sound)

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What you must know | Toolkit | 1. Representing images

2. Representing sound | 3. File size calculations | Worked examples

Quick reference | Typical pitfalls

AQA GCSE Computer Science — **Data Representation: Images and Sound**. This topic covers how bitmap images and digital sound are represented in binary, including pixels, colour depth, sampling rate, and sample resolution. This content is assessed in **Paper 2 (Component 2): Computing concepts**.

Exam questions typically ask you to **describe** how images and sound are represented digitally, **explain** how factors affect quality and file size, and **calculate** file sizes for images and sound. Expect questions worth 2—6 marks requiring both understanding and calculation skills.

Why representing media matters

Computers store everything as binary (0s and 1s), including images and sound. Understanding how media is represented digitally helps you:

- **Understand file sizes:** Why some image files are larger than others, and how to estimate storage requirements
- **Make quality trade-offs:** Higher quality means larger files — knowing the factors helps you make informed decisions
- **Work with digital media:** Photography, music production, video editing, and web design all rely on these concepts

- **Appreciate compression:** Understanding uncompressed file sizes explains why compression is so important

The AQA specification requires you to understand how bitmaps and digital sound work, and be able to calculate file sizes using the formulas provided.

What you must know

This checklist covers the key concepts for representing images and sound. You should be able to explain each point clearly and perform the calculations.

Representing images

- **Understand** what a pixel is and how pixels relate to an image.
- **Describe** image size (width × height in pixels).
- **Describe** colour depth (bits per pixel).
- **Describe** how a bitmap represents an image using pixels and colour depth.
- **Explain** how the number of pixels and colour depth affect file size.
- **Calculate** bitmap image file sizes.
- **Convert** binary data into a bitmap image.
- **Convert** a bitmap image into binary data.

Representing sound

- **Understand** that sound is analogue and must be converted to digital.
- **Understand** that analogue sound is sampled to create digital sound.
- **Describe** sampling rate (samples per second, measured in Hz).
- **Describe** sample resolution (bits per sample).
- **Explain** how sampling rate and sample resolution affect quality and file size.
- **Calculate** sound file sizes.

> **Specification note.** For image file size calculations, AQA states: "Students only need to use colour depth and number of pixels within their calculations." You do not need to account for metadata or file headers — just use the formula provided.

Toolkit

Key formulas

Image file size

Size (bits) = W × H × D

Size (bytes) = (W × H × D) ÷ 8

Where:

- W = image width (pixels)
- H = image height (pixels)
- D = colour depth (bits per pixel)

Sound file size

Size (bits) = rate × res × secs

Size (bytes) = (rate × res × secs) ÷ 8

Where:

- rate = sampling rate (samples per second)
- res = sample resolution (bits per sample)

- secs = duration (seconds)

Key definitions at a glance

Term	Definition	Effect on file size
Pixel	Picture element — a single point/square in an image	More pixels = larger file
Image size	Dimensions in pixels (width × height)	Larger dimensions = larger file
Colour depth	Number of bits used to represent each pixel	Higher colour depth = larger file
Sampling rate	Number of samples taken per second (Hz)	Higher rate = larger file
Sample resolution	Number of bits per sample (bit depth)	Higher resolution = larger file

Common colour depths

Colour depth	Number of colours	Use case
1-bit	2 (black and white)	Simple line drawings, text
4-bit	16	Simple graphics
8-bit	256	Web graphics, GIFs
16-bit	65,536	High colour displays
24-bit	16,777,216 (16.7 million)	True colour photographs

Formula: Number of colours = $2^{\text{colour depth}}$

Unit conversions

Conversion	Value
8 bits	1 byte
1,000 bytes	1 kilobyte (kB)

Conversion	Value
1,000 kilobytes	1 megabyte (MB)
1,000 megabytes	1 gigabyte (GB)

AQA uses decimal prefixes: 1 kB = 1,000 bytes (not 1,024).

Exam tip: Always show your working in calculations. Write out the formula, substitute the values, then calculate step by step. This earns method marks even if you make an arithmetic error.

1. Representing images

Computers represent images as a grid of tiny coloured squares called **pixels**. This type of image is called a **bitmap**. Each pixel stores a colour value as a binary number.

1.1 What is a pixel?

A **pixel** (short for "picture element") is a single point in an image. It is the smallest unit that makes up a bitmap image. When you zoom in on a digital photograph, you can see the individual pixels as tiny squares.

Each pixel has a single colour. When viewed together, thousands or millions of pixels create the illusion of a continuous image. The more pixels an image has, the more detail it can show.

1.2 Image size (resolution)

The **image size** (sometimes called resolution) is the dimensions of an image measured in pixels. It is expressed as **width × height**.

For example, an image that is 1920 × 1080 pixels is 1920 pixels wide and 1080 pixels tall. The total number of pixels is:

1920 × 1080 = 2,073,600 pixels

This is often described as "2 megapixels" (approximately 2 million pixels).

Effect on quality: More pixels means more detail and a sharper image. However, more pixels also means a larger file size.

1.3 Colour depth

Colour depth (also called bit depth) is the number of bits used to represent the colour of each pixel. More bits means more possible colours.

The number of colours available is calculated as:

$$\textbf{Number of colours} = 2^{\textbf{colour depth}}$$

For example:

- 1-bit colour depth: $2^1 = 2$ colours (black and white)
- 8-bit colour depth: $2^8 = 256$ colours
- 24-bit colour depth: $2^{24} = 16{,}777{,}216$ colours

Effect on quality: Higher colour depth allows more colours, producing smoother gradients and more realistic images. However, higher colour depth increases file size.

1.4 How bitmaps work

A bitmap stores the colour value of every pixel in the image. The file contains:

- The colour data for each pixel, row by row
- Information about the image dimensions
- Information about the colour depth

For a simple 1-bit (black and white) image, each pixel is stored as either 0 (white) or 1 (black).

For colour images, the colour depth determines how much data is stored for each pixel. With 24-bit colour (also called "true colour"), each pixel uses 24 bits: 8 bits for red, 8 bits for green, and 8 bits for blue. This RGB model can represent over 16 million different colours by mixing different amounts of red, green, and blue light.

Understanding colour representation

In 24-bit colour images, each pixel's colour is stored using three values (one for each primary colour of light):

Component	Bits	Range	Example
Red	8 bits	0—255	255 = maximum red
Green	8 bits	0—255	255 = maximum green
Blue	8 bits	0—255	255 = maximum blue

Common colours: Black = (0, 0, 0), White = (255, 255, 255), Red = (255, 0, 0), Yellow = (255, 255, 0)

Example: A simple 4×4 bitmap

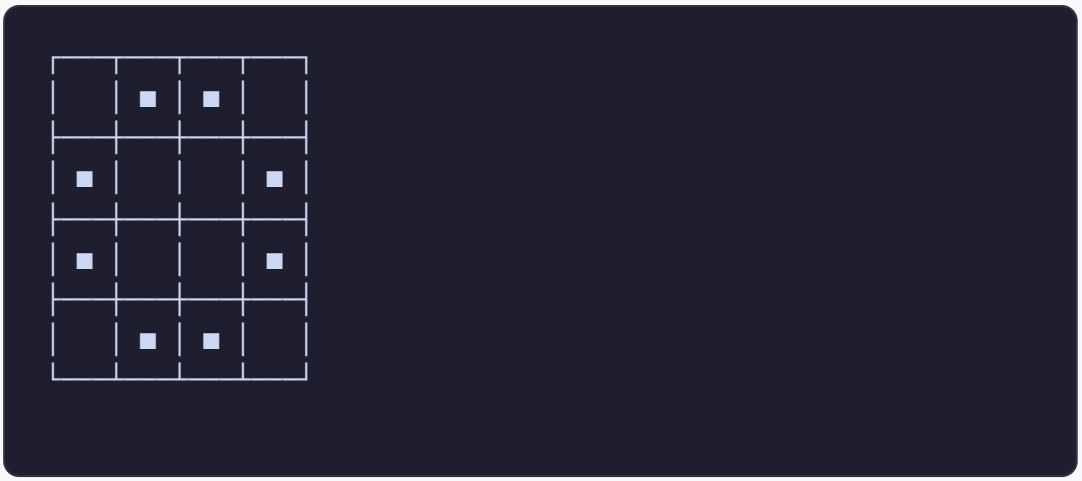
Consider a simple 4×4 pixel black-and-white image with 1-bit colour depth:

Binary representation:

```
Row 1:  0  1  1  0
Row 2:  1  0  0  1
Row 3:  1  0  0  1
Row 4:  0  1  1  0
```

0 = white pixel, 1 = black pixel

Resulting image:



This creates a simple diamond shape.

File size: 4 × 4 × 1 = 16 bits = 2 bytes

1.5 Calculating image file size

The file size of an uncompressed bitmap can be calculated using:

File size (bits) = width × height × colour depth
File size (bytes) = (width × height × colour depth) ÷ 8

Example: Calculate the file size of a 1920 × 1080 image with 24-bit colour depth.

- File size (bits) = 1920 × 1080 × 24 = 49,766,400 bits
- File size (bytes) = 49,766,400 ÷ 8 = 6,220,800 bytes
- File size (MB) = 6,220,800 ÷ 1,000,000 = 6.22 MB

1.6 Quality vs file size trade-offs

Factor	Increase	Effect on quality	Effect on file size
Number of pixels	More pixels (larger dimensions)	More detail, sharper image	Larger file size
Colour depth	More bits per pixel	More colours, smoother gradients	Larger file size

When choosing image settings, there is always a trade-off between quality and file size. High-quality images for printing need more pixels and higher colour depth, while images for web pages may use lower settings to reduce loading times.

Worked Example 1 — Defining a pixel

Question: Define the term "pixel." [2 marks]

Model answer:

Pixel is short for picture element **(1)**. It is a single point/square in a bitmap image **(1)**.

Exam tip: The word "pixel" comes from "picture element" — knowing this shows understanding and often gains a mark.

Worked Example 2 – Explaining colour depth

Question: A bitmap image uses 8-bit colour depth. How many different colours can be represented? Explain your answer. [3 marks]

Model answer:
Number of colours = $2^{\text{colour depth}}$ **(1)**
Number of colours = 2^8 = 256 **(1)**
8 bits can represent 256 different binary patterns, so 256 different colours can be stored **(1)**.

Exam tip: Always show the formula, then substitute the value. This secures method marks even if you miscalculate.

Worked Example 3 – Image file size calculation

Question: Calculate the file size in kilobytes of a bitmap image that is 800 × 600 pixels with 24-bit colour depth. Show your working. [4 marks]

Model answer:
File size (bits) = width × height × colour depth **(1)**
File size (bits) = 800 × 600 × 24 = 11,520,000 bits **(1)**
File size (bytes) = 11,520,000 ÷ 8 = 1,440,000 bytes **(1)**
File size (kB) = 1,440,000 ÷ 1,000 = 1,440 kB **(1)**

Exam tip: Convert units step by step. First calculate bits, then bytes (÷8), then kB (÷1,000) or MB (÷1,000,000).

Worked Example 4 – Binary to bitmap conversion

Question: The following binary data represents a 4×4 bitmap image with 1-bit colour depth where 0 = white and 1 = black. Draw the resulting image. [2 marks]

```
1111 1001 1001 1111
```

Model answer:
Reading each group of 4 bits as a row:

```
1111 → ■ ■ ■ ■
1001 → ■     ■
1001 → ■     ■
1111 → ■ ■ ■ ■

Result:
┌───┬───┬───┬───┐
│ ■ │ ■ │ ■ │ ■ │
├───┼───┼───┼───┤
│ ■ │   │   │ ■ │
├───┼───┼───┼───┤
│ ■ │   │   │ ■ │
├───┼───┼───┼───┤
│ ■ │ ■ │ ■ │ ■ │
└───┴───┴───┴───┘
```

This creates a square/box shape **(2 marks for correct image)**.

> **Exam tip:** Read the binary left to right, top to bottom. Each row has the same number of bits as the image width.

2. Representing sound

Sound in the real world is **analogue** — it varies continuously as pressure waves in the air. Computers can only store **digital** data (discrete binary values). To store sound digitally, the analogue signal must be converted through a process called **sampling**.

2.1 Analogue vs digital

Analogue sound is a continuous wave that varies smoothly over time. When you speak or play music, the sound pressure changes continuously.

Digital sound is a series of discrete measurements (samples) taken at regular intervals. Each sample is stored as a binary number representing the amplitude (loudness) at that moment.

The conversion from analogue to digital is performed by an **analogue-to-digital converter (ADC)**. The reverse conversion (digital to analogue) is performed by a **digital-to-analogue converter (DAC)** when playing back the sound.

2.2 Sampling

Sampling is the process of measuring the amplitude (height/loudness) of an analogue sound wave at regular intervals. Each measurement is called a **sample**.

A sample is a measure of amplitude at a point in time. The sample value is converted to a binary number and stored.

2.3 Sampling rate

The **sampling rate** is the number of samples taken per second, measured in **Hertz (Hz)**. One Hertz means one sample per second.

Common sampling rates:

- **8,000 Hz** — telephone quality (adequate for speech)
- **22,050 Hz** — radio quality (good for voice and simple music)
- **44,100 Hz** — CD quality (standard for music)
- **48,000 Hz** — DVD/professional audio

Why 44,100 Hz for CDs? Human hearing can detect frequencies up to about 20,000 Hz. To accurately capture sounds at a given frequency, you need to sample at least twice that rate. The CD standard of 44,100 Hz samples more than twice the highest frequency humans can hear, ensuring all audible sounds are captured accurately.

Effect on quality: A higher sampling rate captures more detail of the original sound wave, producing better quality audio. However, it also increases file size proportionally — doubling the sampling rate doubles the file size.

2.4 Sample resolution (bit depth)

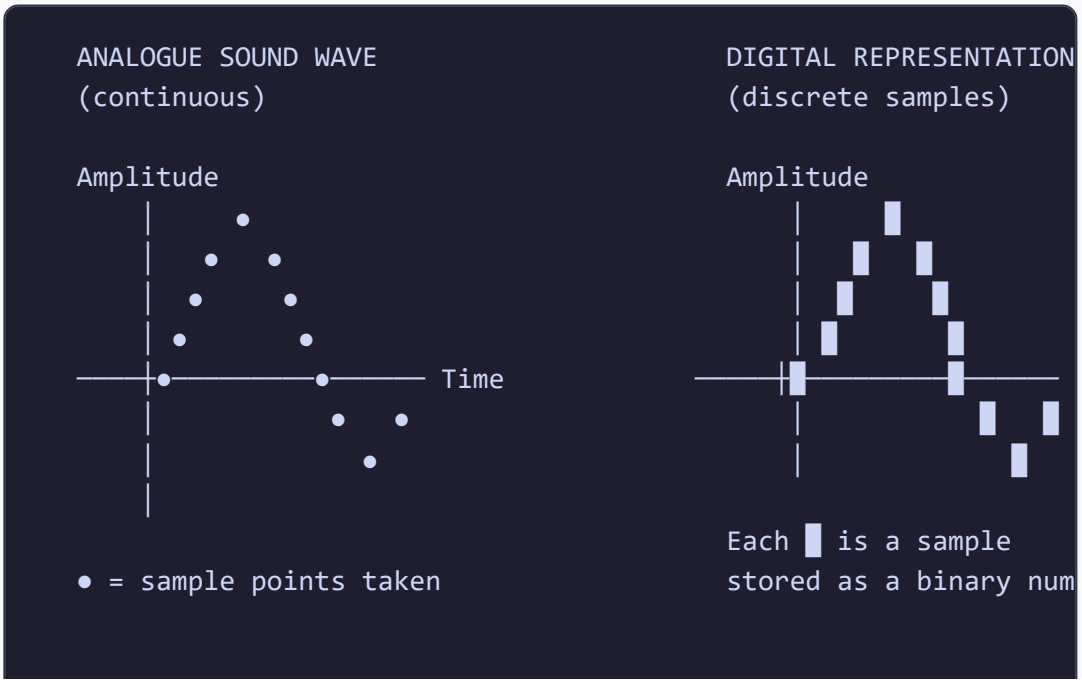
Sample resolution (also called bit depth) is the number of bits used to store each sample. More bits means more precise measurements.

Common sample resolutions:

- **8-bit** — 256 possible amplitude levels
- **16-bit** — 65,536 possible amplitude levels (CD quality)
- **24-bit** — 16,777,216 possible levels (professional)

Effect on quality: Higher sample resolution captures more precise amplitude values, reducing quantisation noise and producing smoother, more accurate sound reproduction.

How sampling works



```
        at regular intervals
```

The analogue wave is measured at regular intervals (determined by the sampling rate). Each measurement becomes a binary value with precision determined by the sample resolution. More samples per second (higher sampling rate) captures more detail of the original wave.

2.5 Quality vs file size trade-offs

Factor	Increase	Effect on quality	Effect on file size
Sampling rate	More samples per second	Captures more detail, especially high frequencies	Larger file size
Sample resolution	More bits per sample	More precise amplitude, less noise	Larger file size
Duration	Longer recording	No effect on quality	Larger file size

Practical example: Comparing audio quality settings

Here's how different quality settings affect a 60-second recording:

Quality	Sampling rate	Resolution	File size	Use case
Telephone	8,000 Hz	8-bit	480 kB	Voice calls
Radio	22,050 Hz	16-bit	2,646 kB	Podcasts, streaming
CD quality	44,100 Hz	16-bit	5,292 kB	Music albums
Professional	48,000 Hz	24-bit	8,640 kB	Studio recording

Note: These are uncompressed (raw) file sizes. Real audio files are usually compressed to reduce size while maintaining acceptable quality.

Worked Example 5 — Explaining sampling

Question: Explain why sound must be sampled before it can be stored on a computer. [2 marks]

Model answer:

Sound is analogue/continuous and computers can only store digital/binary data **(1)**. Sampling measures the amplitude of the sound wave at regular intervals, converting it to digital values that can be stored **(1)**.

Exam tip: Always mention the analogue-to-digital conversion — this is the key concept. Sampling is the method of making this conversion.

Worked Example 6 – Sampling rate and quality

Question: Explain how increasing the sampling rate affects the quality and file size of a sound recording. [4 marks]

Model answer:
Quality: A higher sampling rate means more samples are taken per second **(1)**. This captures more detail of the original sound wave, producing a more accurate representation of the original sound **(1)**.
File size: More samples means more data to store **(1)**. The file size increases proportionally with the sampling rate **(1)**.

Exam tip: For "effect on quality and file size" questions, always address both aspects. Structure your answer clearly with separate paragraphs for each.

Worked Example 7 – Comparing sample resolutions

Question: A sound is recorded twice: once with 8-bit sample resolution and once with 16-bit sample resolution. Compare the two recordings in terms of quality. [3 marks]

Model answer:
The 8-bit recording can represent 2^8 = 256 different amplitude levels **(1)**. The 16-bit recording can represent 2^{16} = 65,536 different amplitude levels **(1)**. The 16-bit recording will be higher quality because it can capture more precise amplitude values, resulting in smoother, more accurate sound reproduction **(1)**.

Exam tip: When comparing, give specific numbers (256 vs 65,536 levels). This demonstrates calculation skills and understanding.

Worked Example 8 – Defining key terms

Question: Define the following terms: (a) sampling rate, (b) sample resolution. [4 marks]

Model answer:
(a) **Sampling rate:** The number of samples taken per second **(1)**, measured in Hertz (Hz) **(1)**.
(b) **Sample resolution:** The number of bits used to store each sample **(1)**. It determines the number of possible amplitude levels that can be recorded **(1)**.

Exam tip: Learn these definitions precisely. Include the units (Hz for sampling rate, bits for resolution) for full marks.

3. File size calculations

Being able to calculate file sizes is a key exam skill. This section provides step-by-step methods for both image and sound file size calculations.

3.1 Image file size calculation method

Step-by-step method

1. **Identify the values:** width (pixels), height (pixels), colour depth (bits)
2. **Calculate total pixels:** width × height
3. **Calculate size in bits:** total pixels × colour depth
4. **Convert to bytes:** divide by 8
5. **Convert to larger units if needed:** divide by 1,000 for kB, by 1,000,000 for MB

3.2 Sound file size calculation method

Step-by-step method

1. **Identify the values:** sampling rate (Hz), sample resolution (bits), duration (seconds)
2. **Calculate size in bits:** sampling rate × sample resolution × duration
3. **Convert to bytes:** divide by 8
4. **Convert to larger units if needed:** divide by 1,000 for kB, by 1,000,000 for MB

Converting time: If duration is given in minutes, multiply by 60 to get seconds. If given in hours, multiply by 3,600.

3.3 Common calculation scenarios

Scenario	Example	Key consideration
HD image	1920 × 1080, 24-bit	Large pixel count × high colour depth = big file
Simple icon	32 × 32, 8-bit	Small dimensions keep file small

Scenario	Example	Key consideration
CD quality audio	44,100 Hz, 16-bit, 180 seconds	Convert minutes to seconds first
Voice recording	8,000 Hz, 8-bit, 60 seconds	Lower rates for speech = smaller files

3.4 Reverse calculations

Sometimes exam questions ask you to work backwards — for example, finding the duration of a recording given the file size, or finding the colour depth of an image.

Rearranged formulas

To find colour depth:

Colour depth = File size (bits) ÷ (width × height)

To find duration:

Duration (seconds) = File size (bits) ÷ (sampling rate × sample resolution)

To find sampling rate:

Sampling rate = File size (bits) ÷ (sample resolution × duration)

Worked Example 9 — Sound file size calculation

Question: Calculate the file size in megabytes of a 3-minute audio recording with a sampling rate of 44,100 Hz and 16-bit sample resolution. Show your working. [4 marks]

Model answer:
Duration in seconds = 3 × 60 = 180 seconds **(1)**
File size (bits) = sampling rate × sample resolution × duration
File size (bits) = 44,100 × 16 × 180 = 127,008,000 bits **(1)**
File size (bytes) = 127,008,000 ÷ 8 = 15,876,000 bytes **(1)**
File size (MB) = 15,876,000 ÷ 1,000,000 = 15.876 MB ≈ 15.9 MB **(1)**

Exam tip: Don't forget to convert minutes to seconds! This is a common error that loses marks.

Worked Example 10 — Comparing file sizes

Question: Two images have the same dimensions (640 × 480). Image A uses 8-bit colour depth and Image B uses 24-bit colour depth. Calculate the file size of each image and explain the difference. [5 marks]

Model answer:
Image A (8-bit):

File size = 640 × 480 × 8 = 2,457,600 bits = 307,200 bytes = 307.2 kB **(1)**
Image B (24-bit):
File size = 640 × 480 × 24 = 7,372,800 bits = 921,600 bytes = 921.6 kB **(1)**
Comparison:
Image B is 3 times larger than Image A **(1)**. This is because 24-bit colour depth uses 3 times as many bits per pixel as 8-bit colour depth **(1)**. Image B can represent 16.7 million colours compared to only 256 colours in Image A, but requires more storage space **(1)**.

Exam tip: When comparing, calculate the ratio (24÷8=3) to explain the relationship between the file sizes clearly.

Worked Example 11 — Reverse calculation (finding duration)

Question: A sound file is 3,528,000 bytes. It was recorded with a sampling rate of 44,100 Hz and 8-bit sample resolution. Calculate the duration of the recording in seconds. [3 marks]

Model answer:
File size (bits) = 3,528,000 × 8 = 28,224,000 bits **(1)**
Duration = File size (bits) ÷ (sampling rate × sample resolution) **(1)**
Duration = 28,224,000 ÷ (44,100 × 8) = 28,224,000 ÷ 352,800 = 80 seconds **(1)**

Exam tip: In reverse calculations, you may need to convert bytes to bits first (multiply by 8) before using the formula.

Worked Example 12 — Extended calculation with comparison

Question: A company needs to store 100 photographs. Each photograph is 3000 × 2000 pixels with 24-bit colour depth.
(a) Calculate the total storage needed in gigabytes. [4 marks]
(b) The company decides to reduce the colour depth to 8-bit. How much storage would they save? [2 marks]

Model answer:
(a) Original storage:
Size per image (bits) = 3000 × 2000 × 24 = 144,000,000 bits **(1)**
Size per image (bytes) = 144,000,000 ÷ 8 = 18,000,000 bytes **(1)**
Total for 100 images = 18,000,000 × 100 = 1,800,000,000 bytes **(1)**
Total in GB = 1,800,000,000 ÷ 1,000,000,000 = 1.8 GB **(1)**
(b) Reduced storage:

With 8-bit colour depth, file size is 8÷24 = 1/3 of original
New total = 1.8 ÷ 3 = 0.6 GB **(1)**
Storage saved = 1.8 − 0.6 = 1.2 GB **(1)**

Exam tip: For multi-part questions, use your answer from part (a) in part (b). If you made an error in (a), you can still get full marks in (b) if your method is correct.

Worked examples summary

The following table summarises the 12 worked examples covered in this topic, organised by section and exam skill.

Example	Focus	Marks	Key skill
1	Defining a pixel	2	Clear definition with etymology
2	Explaining colour depth	3	Formula application (2^n)
3	Image file size calculation	4	Step-by-step calculation
4	Binary to bitmap conversion	2	Interpreting binary as image
5	Explaining sampling	2	Analogue to digital conversion
6	Sampling rate and quality	4	Quality vs file size trade-off
7	Comparing sample resolutions	3	Numerical comparison
8	Defining key terms	4	Precise definitions with units
9	Sound file size calculation	4	Time conversion + calculation
10	Comparing file sizes	5	Calculate and explain ratio
11	Reverse calculation	3	Rearranging formulas
12	Extended multi-part calculation	6	Real-world application

Quick reference

Key definitions (mark-scheme style)

Term	Definition
Pixel	Picture element; a single point in a bitmap image
Image size	The dimensions of an image measured in pixels (width × height)
Colour depth	The number of bits used to represent each pixel

Term	Definition
Bitmap	An image made up of a grid of pixels, each storing a colour value
Sample	A measurement of amplitude at a point in time
Sampling rate	The number of samples taken per second, measured in Hertz (Hz)
Sample resolution	The number of bits used to store each sample (bit depth)
Analogue	A continuous signal that varies smoothly over time
Digital	Data represented as discrete binary values

Formulas

Image file size

Size (bits) = W × H × D
Size (bytes) = (W × H × D) ÷ 8
W = width, H = height, D = colour depth

Sound file size

Size (bits) = rate × res × secs
Size (bytes) = (rate × res × secs) ÷ 8
rate = sampling rate, res = resolution, secs = duration

Colour depth reference

Colour depth	Calculation	Number of colours
1-bit	2^1	2
4-bit	2^4	16
8-bit	2^8	256
16-bit	2^{16}	65,536
24-bit	2^{24}	16,777,216

Unit conversions

From	To	Operation
bits	bytes	÷ 8

From	To	Operation
bytes	kilobytes (kB)	÷ 1,000
kilobytes	megabytes (MB)	÷ 1,000
megabytes	gigabytes (GB)	÷ 1,000
minutes	seconds	× 60

Typical pitfalls

Pitfall 1 — Forgetting to convert minutes to seconds.
Wrong: Using 3 minutes directly in the sound formula.
Right: Convert first: 3 minutes = 3 × 60 = 180 seconds.

Pitfall 2 — Confusing bits and bytes.
Wrong: Forgetting to divide by 8 when converting to bytes.
Right: Always state units clearly and remember: 8 bits = 1 byte.

Pitfall 3 — Using 1024 instead of 1000.
Wrong: "1 kB = 1024 bytes"
Right: AQA uses decimal prefixes: 1 kB = 1,000 bytes.

Pitfall 4 — Confusing image size and colour depth.
Wrong: "Higher resolution means more colours."
Right: Resolution (pixels) affects detail; colour depth (bits per pixel) affects number of colours.

Pitfall 5 — Not showing working in calculations.
Wrong: Just writing the final answer.
Right: Show formula, substitution, and each step to earn method marks.

Pitfall 6 — Confusing sampling rate and sample resolution.
Wrong: "Higher sampling rate means more bits per sample."
Right: Sampling rate = samples per second (Hz). Sample resolution = bits per sample.

Pitfall 7 — Forgetting Hz means "per second."
Wrong: Not understanding what 44,100 Hz means.
Right: 44,100 Hz = 44,100 samples taken every second.

Pitfall 8 — Incomplete explanations of quality.
Wrong: "Higher colour depth is better quality."
Right: "Higher colour depth means more colours can be represented, producing smoother gradients and more realistic images."

Pitfall 9 — Wrong direction in binary-to-image conversion.
Wrong: Reading binary right-to-left or bottom-to-top.
Right: Read left-to-right, top-to-bottom, matching the pixel grid.

Pitfall 10 — Forgetting to multiply for multiple files.
Wrong: Calculating size for one image when asked about 100 images.
Right: Read the question carefully — multiply by the number of files if needed.

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